LESSON 309 (1.5 AATD)

Lesson Objective
This lesson continues to develop the student's knowledge and skill with respect to attitude instrument flying. Also, VOR, DME arc, and NDB navigation procedures will be reinforced.

Briefing
- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

Lesson Content (Simulator)
System Power Up

- Student Tells, Student Does.
    - Have the student start the simulator using the checklist and assist.
    - Ensure that the student checks GPS database currency.
        - Half-left, half-up, flag out of view.
    - Ensure that there are no red 'x's.
    - Have the student start the engine using the checklist.

Instrument Cockpit Check

- Student Tells, Student Does.
    - Have the student read each item from the checklist out loud.
    - Demonstrate to the student what needs to be done for each item.
    - Discipline the student to set bugs, bearing pointers, and any auxiliary features.

Before Taxi Checks

- Student Tells, Student Does.
    - Airspeed indicator (PFD and standby): zero.
    - Attitude indicator (PFD): pitch/bank deviations +/- 5 degrees.
    - Standby attitude indicator: discuss standby battery check and adjust the miniature aircraft (cannot be performed in the ATD).
    - Altimeter (PFD and standby): +/- 75 feet of station elevation.
    - Vertical speed indicator (PFD): reads 0 or use current indication.
    - Trim coordinator: wings level or no vector when still; inclinometer centered when still; skids when turning.
    - HSI: matches compass and turns appropriately.
    - Magnetic compass: indicates heading properly, no bubbles, card present, and swings freely.

Give the student a departure via radar vectors. The instructor simulates ATC. Have the student switch to the appropriate frequencies throughout the simulator lesson.

VOR Intercepting and Tracking

- Ensure that the student can TUNE and ID the VOR.
- Intercepting and Tracking Within 5NM:
  - Instructor Tells, Student Does.
    - Specify a radial for the student to intercept outbound.
    - Have the student make note of the distance.
    - The proper radial is to be tuned before the intercept is applied with a FROM indication.
    - Ensure that the student uses the heading bug.
    - Give the student crosswind to see how well they can track a radial.
- Intercepting and Tracking Outside 5NM:
  - Instructor Tells, Student Does.
    - Give the student another radial to intercept outbound from the VOR.
    - Listen for the student to note the current radial.
    - Ensure that the student notes the difference and doubles.
    - Ensure that the student is twisting the desired course before applying any intercept.
    - Ask the student which direction the airplane should fly to intercept – the shortest turn should be used.
    - Guard the student against using <20° or >90°.
    - Have the student bug the intercept heading and intercept the radial.
    - Repeat the above steps as time permits, challenging the student with Inbound to Outbound or Outbound to Inbound turns.

Intercepting and Tracking DME Arcs

- Intercepting and Tracking from a Radial:
  - Instructor Tells, Student Does.
    - Note the radial, direction, and distance from the station.
    - Give ARC Instructions with Compass Direction, Distance, and the Fix.
    - Make sure the student understands the instruction.
    - The no-wind intercept is 100° to the present radial (Advance OBS 10° and add 90° in direction of the arc).
    - Join the arc at 0.5 nm prior to the arc (depending on headwind/tailwind) – immediately advance OBS 10°
    - Make sure OBS is set up in the FROM indication.
    - If undershooting the arc, roll out early; If overshooting, continue to turn to intercept the arc.
    - Let the student choose what method they want to use to track DME arcs.
- Intercepting and Tracking a DME Arc from Radar Vectors:
  - Instructor Tells, Student Does.
    - Tell the student to program a Non-VLOC waypoint into the GPS for the DME arc.

- Vector the student to intercept a DME Arc outside a Final Approach Fix of a back-course approach.
- Pause if needed – Give ARC Instructions with Compass Direction, Distance, and the Fix.
- This will be a DME arc to join a back course outbound.
- Ensure that the localizer is Tuned, Identified, and Twisted on Nav #1.
- GPS #2 will be used for the arc on the bearing pointer.
- At 1.5 DME prior to the arc, note the radial, and calculate the intercept heading.
- The no-wind intercept is 100° to the present radial (or 90° to the succeeding radial).
- Join the arc at 0.5 nm prior (depending on degrees to be turned and winds) to the arc or as necessary to prevent overshoot/undershoot.

Localizer Back Course Intercepting and Tracking

- Intercepting the Localizer Back Course:
  - Instructor Tells, Student Does.
    - Choose a localizer, such as I-MLB, to track.
    - Vector the student toward the localizer (simulate how approach would do it).
    - Coach the student through setting up the instruments:
      - Weather (obtain the ATIS).
      - Instruments (load the procedure, tune and ID the localizer, set up the PFD to the correct NAV source, and set the approach course (BE CAREFUL OF REVERSE SENSING, SET THE FRONT COURSE).
      - Radios (set COM1/COM2).
      - Enroute brief (this step can be skipped in stage 1).
      - Descent and arrival checklist.
    - Stress to the student to turn early once the needle leaves full-scale deflection.
    - Add flaps 10 prior to joining the localizer.
- Tracking the Localizer Back Course:
  - Instructor Tells, Student Does.
    - Remind the student of any wind direction indicators or track bugs.
    - Stress the use of early, small corrections.
    - Have the student track inbound until the runway threshold.
    - If the student can, then have them follow the glideslope.
    - If not, then instruct them to do a constant rate descent at 90 knots starting at the final approach fix (tell them where that is).
    - Highlight that the autopilot can track the localizer back course by pressing the REV button on the S-TEC 55X.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 310.
- Assign Homework:
    - FIT procedures for VOR Intercepting and Tracking.
    - FIT procedures for Intercepting and Tracking DME Arcs from radials and radar vectors.

Completion Standards

The student should:

1. Demonstrate both the instrument cross-check and interpretation necessary to maintain control with respect to attitude instrument flying.
2. Demonstrate an understanding of VOR, DME arc, and NDB procedures.